

PracticeMidterm2_VB

Sunday, November 6, 2022 6:32 PM



PracticeMidterm2_VB

Math 340
Practice Exam 2
Time Limit: 75 Minutes

Name (Print): _____
TA's name and Section#: _____
Student ID Number: _____

Key

This exam contains 6 pages (including this cover page) and 8 problems. The last page is the scratch paper. If any pages are missing, please inform the instructor immediately.

Directions:

- Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.
- You may *not* use your calculators, books and notes.
- You must show all your work. A stranger should be able to pick up your paper and follow your reasoning. You must use proper notation and define the events clearly. Partial credits may be given.
- Unless otherwise noted, your answers must be in fraction form or terminating decimals.
- Circle your answer to the True/False problems and justify your answer.
- For the free response problems you need to show all necessary work in the space provided for each problem. Answers must be justified with sufficient work, and partial credit will be given for appropriate work shown.

I do not lie, cheat or steal, or tolerate those who do.

By signing below you indicate that all your work is your own and that you have neither given nor received help from external sources.

Signature: _____

T/F	
P.6	
P.7	
P.8	
Total	

True/False Questions: If true, give your justification. If false, give a counterexample or some reasoning.

1. (2 points) Let W be the set of all vectors of the form $\begin{bmatrix} x \\ y \end{bmatrix}$, where $x + y = 1$. W is a subspace of $\mathbb{R}^2$ with the usual operations of vector addition and scalar multiplication.

A. True.

☒ B. False.

$$2 \underset{\in W}{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \text{ not in } W, \text{ not closed under scalar mult.}$$

2. (2 points) Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ be a spanning set of the vector space $\mathbb{R}^3$. If $\mathbf{v}_1$ is a linear combination of $\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$, then $\{\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is also a spanning set of $\mathbb{R}^3$.

☒ A. True.

B. False.

any linear comb. using $\vec{v}_1$ can use $\vec{v}_2, \vec{v}_3, \vec{v}_4$ instead

3. (2 points) If a linear transformation $L : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is one-to-one, then L is invertible.

☒ A. True.

B. False.

$$\dim \ker L = 0.$$

$$\dim \text{image}(L) = 3$$

4. (2 points) Let $L : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $L([u_1 \ u_2 \ u_3]^T) = [0 \ u_1^2 \ u_2 + u_3]^T$. Then L is a linear transformation.

A. True.

☒ B. False.

$$L\left(2 \underset{u.}{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}\right) = L\left(\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix}$$

$$u. \quad 2L\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = 2 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$$

5. (2 points) Let S and T be two finite nonempty subsets of a vector space such that S is contained in T . If T is linearly dependent, so is S .

A. True.

☒ B. False.

$$T = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\} \leftarrow \text{lin. dep.}$$

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} \leftarrow \text{lin. indep.}$$

6. (10 points) Let $L: \mathbb{R}^5 \rightarrow \mathbb{R}^4$ be defined by

matrix A

$$L \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix} = \begin{bmatrix} 1 & 0 & -1 & 3 & -1 \\ 1 & 1 & 0 & 2 & -1 \\ 2 & 1 & -1 & 5 & -1 \\ 0 & -1 & -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = u_1 \vec{v}_1 + u_2 \vec{v}_2 + u_3 \vec{v}_3 + u_4 \vec{v}_4 + u_5 \vec{v}_5$$

$\vec{v}_1 \quad \vec{v}_2 \quad \vec{v}_3 \quad \vec{v}_4 \quad \vec{v}_5$

- (a) Find a basis for $\text{Ker}(L)$.
 (b) What is the dimension of $\text{Ker}(L)$?
 (c) Find a basis for $\text{range}(L)$ (that is, the same thing as $\text{im}(L)$).
 (d) What is the dimension of $\text{range}(L)$?

(a) Reduce A:

$-r_2 + r_3 \rightarrow r_3, r_2 + r_4 \rightarrow r_4$

$$\begin{bmatrix} 1 & 0 & -1 & 3 & -1 \\ 1 & 1 & 0 & 2 & -1 \\ 2 & 1 & -1 & 5 & -1 \\ 0 & -1 & -1 & 1 & 0 \end{bmatrix} \xrightarrow{\substack{-r_1 + r_2 \rightarrow r_2 \\ -2r_1 + r_3 \rightarrow r_3}} \begin{bmatrix} 1 & 0 & -1 & 3 & -1 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & -1 & 1 \\ 0 & -1 & -1 & 1 & 0 \end{bmatrix} \xrightarrow{\substack{-r_2 + r_3 \rightarrow r_3 \\ r_2 + r_4 \rightarrow r_4}} \begin{bmatrix} 1 & 0 & -1 & 3 & -1 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$r_3 + r_1 \rightarrow r_1$

$\begin{matrix} \text{O=pivot} & & \text{free} \\ x_1 & x_2 & x_3 & x_4 & x_5 \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} & \begin{matrix} \uparrow & \uparrow & & & \uparrow \end{matrix} \end{matrix}$

$x_3 = t \quad x_4 = r \quad x_5 = 0$

$x_1 = t - 3r \quad x_2 = -t + r$

(a) Basis for $\text{ker}(L)$: $\begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

(b) $\dim \text{ker}(L) = 2$

(c) Comes from orig. column vectors $\rightarrow \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix} \right\}$

(d) $\dim \text{image } L = 3$

7. (10 points) Let W be the subspace of $\mathbb{R}^4$ spanned by the set of vectors $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -6 \\ -3 \end{bmatrix}, \begin{bmatrix} -1 \\ -5 \\ 1 \\ 0 \end{bmatrix} \right\}$

Find the basis for W . What is the dimension of W ?

Put in matrix:

$$\begin{array}{ccccc} \vec{v}_1 & \vec{v}_2 & \vec{v}_3 & \vec{v}_4 & \vec{v}_5 \\ \begin{bmatrix} 1 & 0 & 1 & 1 & -1 \\ 1 & 1 & 0 & 1 & -5 \\ 0 & 2 & 1 & -6 & 1 \\ -1 & 1 & -1 & -3 & 0 \end{bmatrix} & \xrightarrow{\text{math...}} & \begin{bmatrix} 1 & 0 & 0 & 3 & -4 \\ 0 & 1 & 0 & -2 & -1 \\ 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

↑ ↑ ↑ ↑ ↑
pivot columns

Choose $\begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}$ for the basis.

$$\dim(W) = 3$$

8. (10 points) Let P_2 denote the vector space of polynomials in t of degree at most 2. For what values of λ are the vectors $\mathbf{v}_1 = 2t^2 + t + 2$, $\mathbf{v}_2 = t^2 + \lambda$ and $\mathbf{v}_3 = t^2 + 2t + \lambda^2$ linearly independent?

↳ see when a_1, a_2, a_3 give

$$a_1 \vec{v}_1 + a_2 \vec{v}_2 + a_3 \vec{v}_3 = \vec{0} \leftarrow \text{0 poly}$$

$$a_1(2t^2 + t + 2) + a_2(t^2 + \lambda) + a_3(t^2 + 2t + \lambda^2) = 0$$

$$0 = \underbrace{(2a_1 + a_2 + a_3)}_{=0} t^2 + \underbrace{(a_1 + 2a_3)}_{=0} t + \underbrace{(2a_1 + \lambda a_2 + a_3 \lambda^2)}_{=0}$$

$$2a_1 + a_2 + a_3 = 0$$

$$a_1 + 0 + 2a_3 = 0$$

$$2a_1 + \lambda a_2 + \lambda^2 a_3 = 0$$

$$\left. \begin{array}{l} 2a_1 + a_2 + a_3 = 0 \\ a_1 + 0 + 2a_3 = 0 \\ 2a_1 + \lambda a_2 + \lambda^2 a_3 = 0 \end{array} \right\} \rightarrow \text{matrix} \begin{bmatrix} 2 & 1 & 1 & | & 0 \\ 1 & 0 & 2 & | & 0 \\ 2 & \lambda & \lambda^2 & | & 0 \end{bmatrix}$$

$= A \rightarrow$ has nontrivial soln when $\det(A) = 0$

$$\det(A) = -1(\lambda^2 - 4) - \lambda(4 - 1)$$

$$= -\lambda^2 + 4 - 3\lambda = 0$$

$$\lambda^2 + 3\lambda - 4 = 0$$

$$(\lambda + 4)(\lambda - 1) = 0$$

$\lambda = -4, 1 \rightarrow$ has nontrivial soln (so lin. dep).

So when $\lambda \neq -4, 1$, these vectors are linearly independent

Scratch Paper